Active Directory Environment Setup Report

1. Introduction

This report presents the implementation of a Windows Active Directory environment including a Domain Controller, DNS, DHCP, IIS server, and Group Policy configurations. The objective is to simulate an organizational network and enforce access control policies based on user roles.

2. Executive Summary

A complete enterprise-like environment was successfully deployed using Windows Server. Users were organized into departments (HR and Technical), and access control policies were applied to restrict unauthorized actions.
Security controls such as disabling system tools, restricting login hours, and blocking access to sensitive resources were implemented.
The system demonstrates a structured and secure configuration aligned with organizational requirements.

3. Technical Report

Component	Details
Domain Name	khateeb.com
Domain Controller (DC) IP	192.168.1.91
Client Workstation IP	192.168.1.100
Operating Systems	Windows Server 2022 & Windows 10 Pro

3.1 Environment Setup

Installed Windows Server → Opened Server Manager → Add Roles and Features → Installed AD DS, DHCP, IIS → Promoted to Domain Controller → Created domain khateeb.com

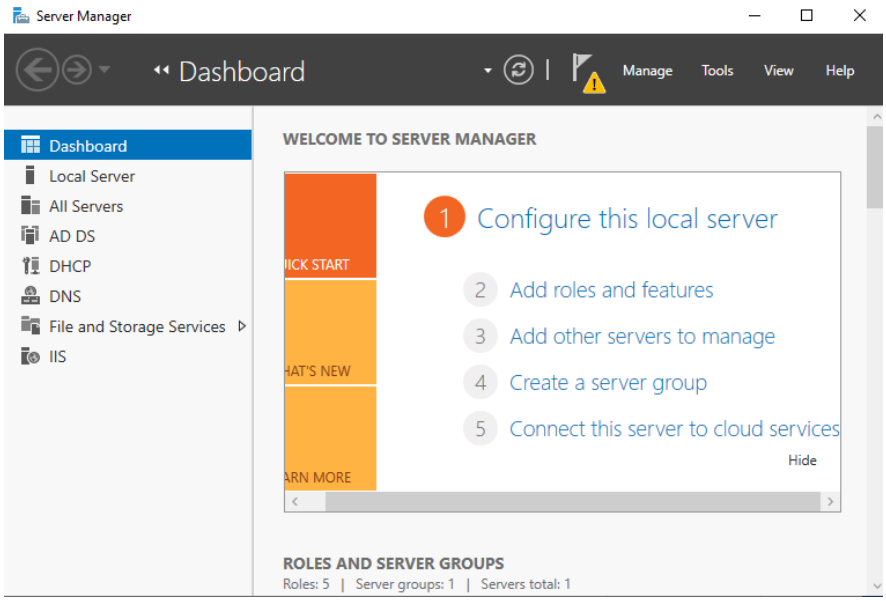


Figure 1: Installed roles and features.

3.2 Domain Configuration

- Verification: Confirmed domain health via ADDS and verified DNS zone creation for khateeb.com.

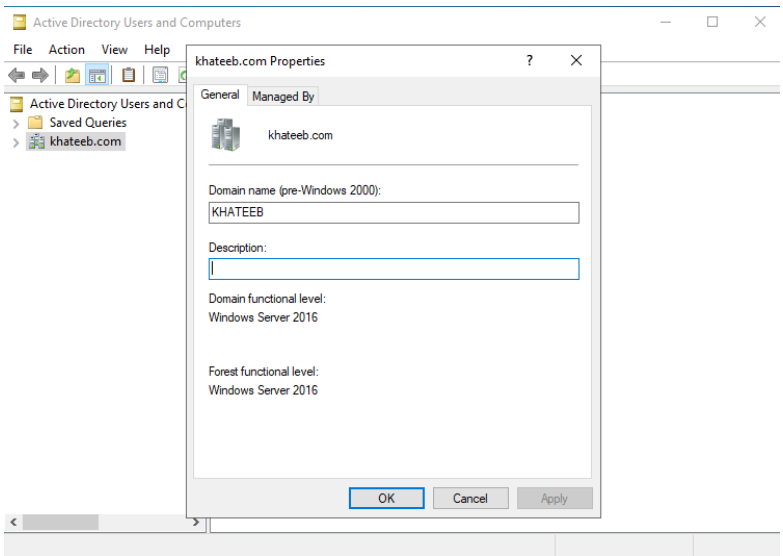


Figure 2: Domain successfully configured.

3.3 Client Join to Domain

Set DNS to server IP → System settings → Join domain khateeb.com → Login with Administrator → Restart

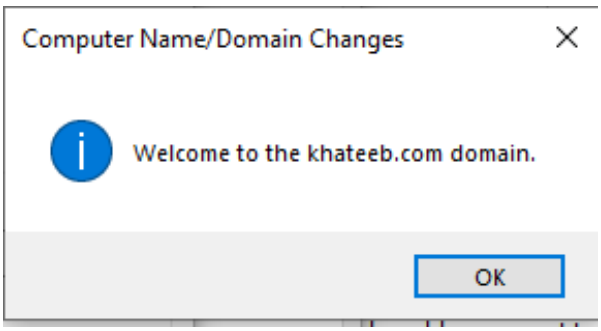


Figure 3: Client joined to domain.

3.4 User and OU Management

Open AD Users and Computers → Create OU (HR, Technical) → Create 4 users → Assign users to Ous

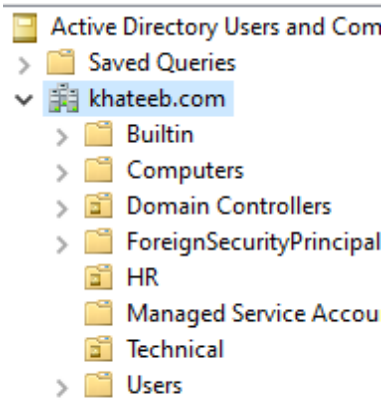


Figure 4: Organizational Units created.

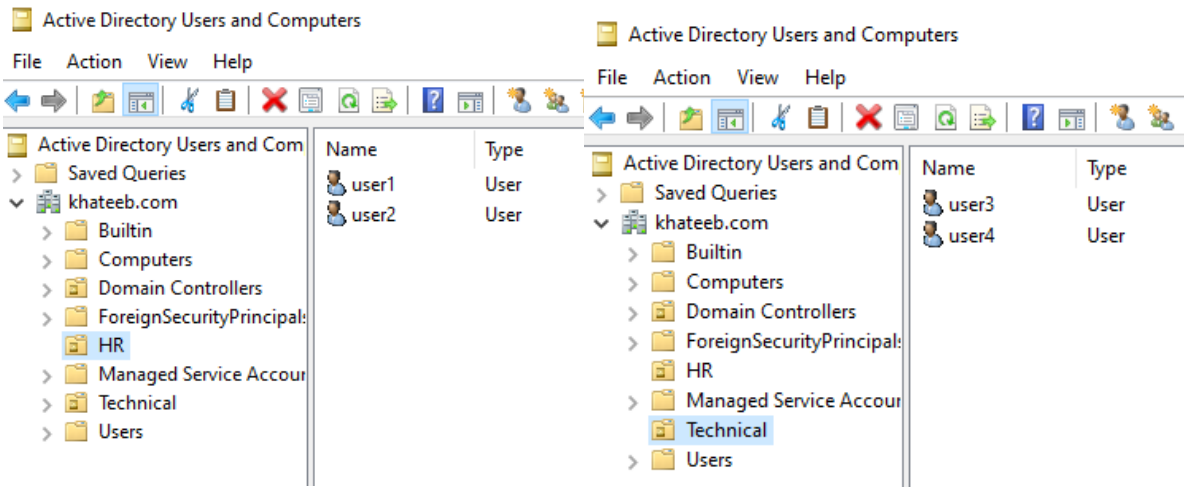


Figure 5: Users assigned to departments.

3.5 IIS Website Hosting

Download template → Extract files → Copy to wwwroot → Open browser using server IP

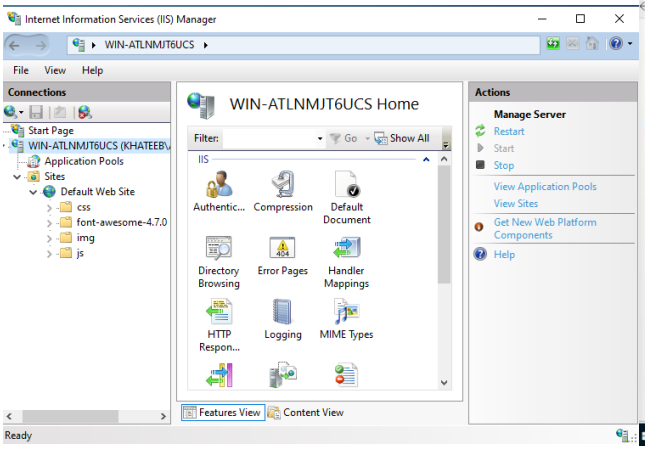


Figure 6: IIS configuration.

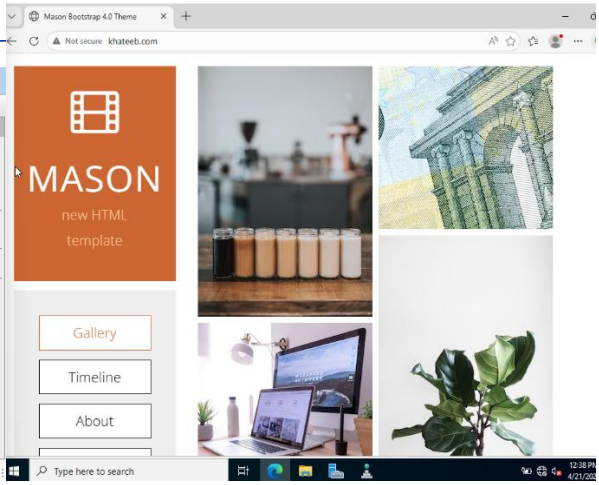


Figure 7: Website successfully hosted.

3.6 Group Policy Configuration

HR Policy

Create GPO (HR) → Disable CMD & PowerShell → Block server IP → Create GPO (Domain) → Disable Control Panel → Disable USB

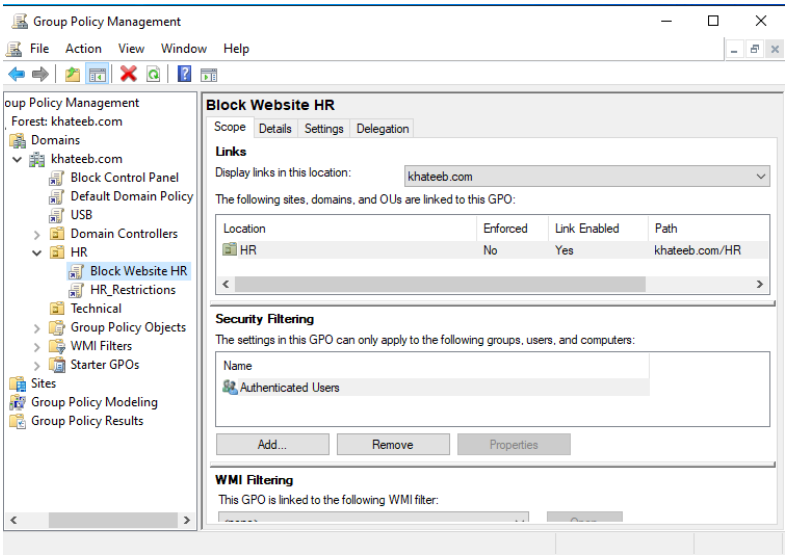


Figure 8: HR policy applied.

General Policy

- Admin Control: Disabled access to the Control Panel.
- DLP Measures: Enforced a USB Access Denied policy across all removable storage.

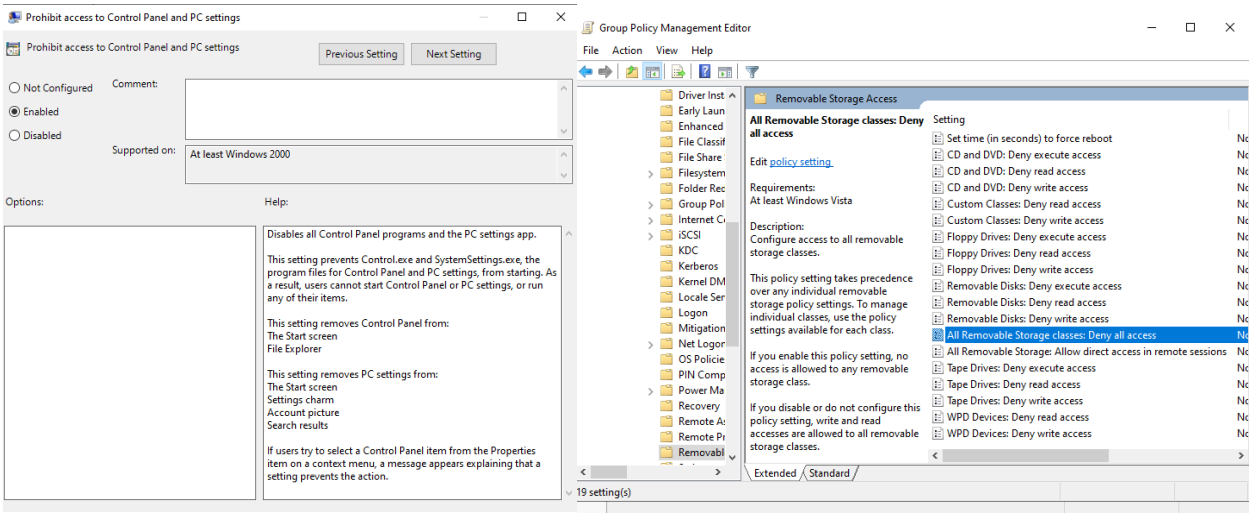


Figure 10: USB access blocked.

3.7 Logon Hours Restriction

User Properties → Account → Logon Hours → Set 8 AM – 5 PM

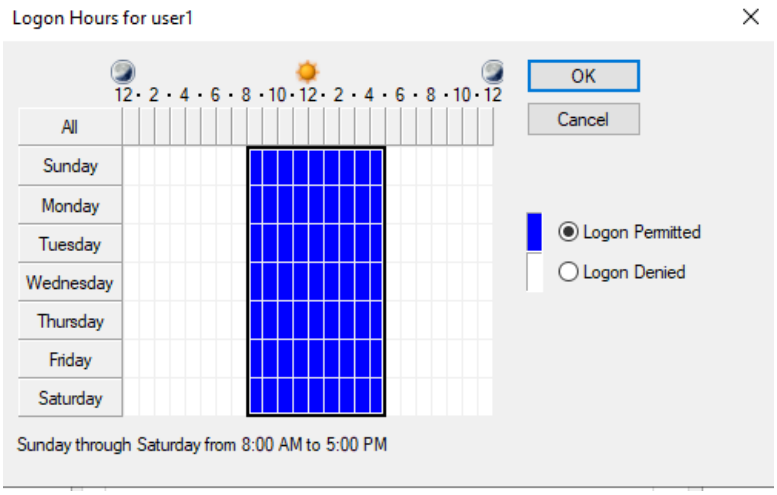


Figure 11: Login time restriction applied.

3.8 Testing and Validation

HR user → CMD blocked & website blocked → Technical user → website accessible → Control Panel blocked

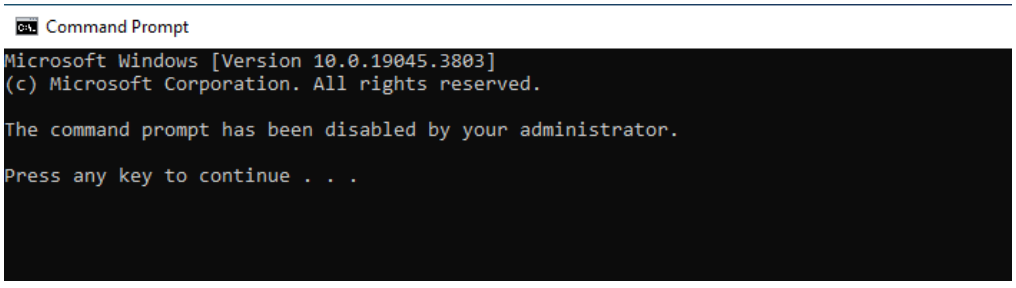


Figure 12: CMD access denied.

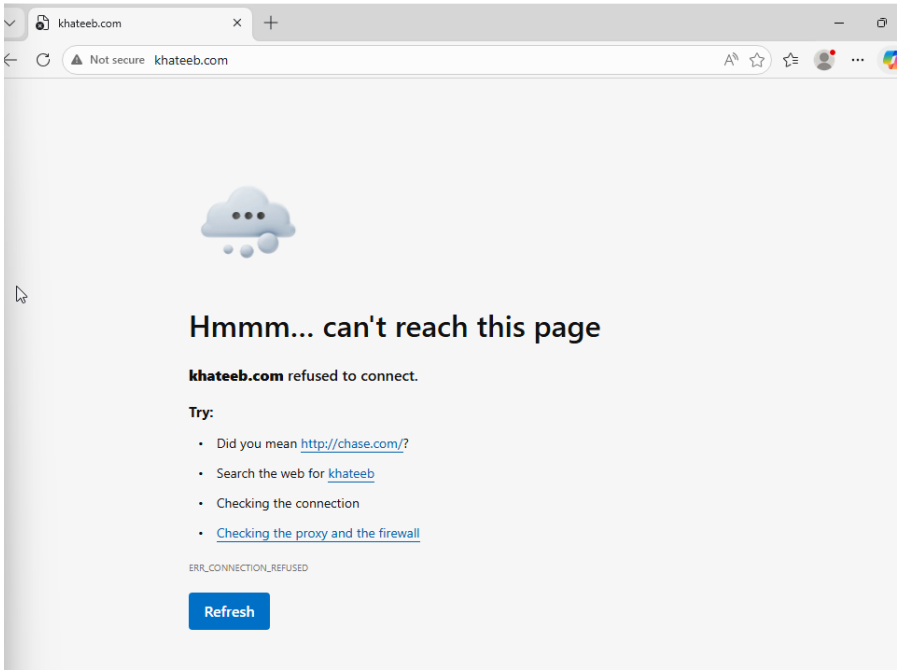


Figure 13: HR user denied website access.

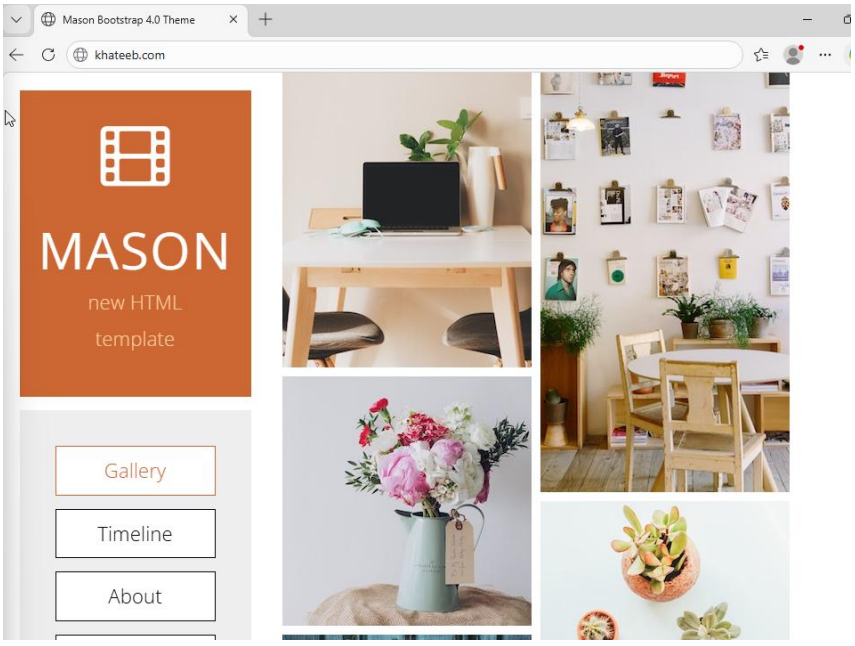


Figure 14: Technical user access granted.

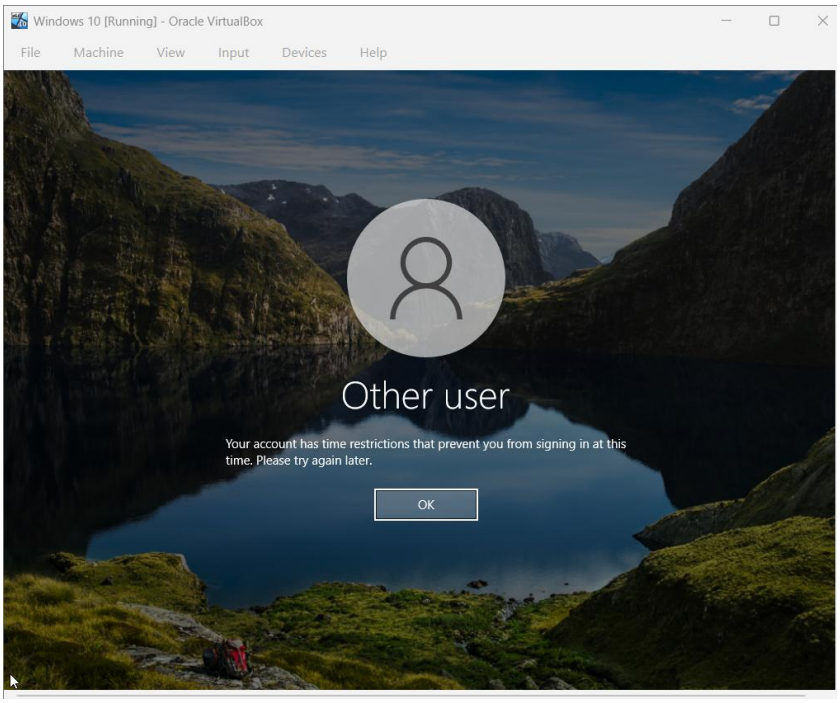


Figure 15: Verification of Logon Hours policy preventing user access outside business hours.

4. Conclusion

The project successfully demonstrated the deployment of a secure Active Directory environment. Policies were applied effectively to control user access and enforce organizational rules. The system reflects a structured and controlled network setup suitable for enterprise environments.